

Quiz 13 Solutions

STAT 109: Introductory Biostatistics

Quiz 13 solutions

Part A

- A1(a): **b**
 - A1(b): **c**
 - A2(a): **a**
 - A2(b): **a**
 - A3(a): **c**
 - A3(b): **a**
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Part B

B1.

$$\widehat{Volume} = -36.943 + 5.066(\text{Girth})$$

B2.

For each additional 1 inch of tree girth, predicted **mean** volume increases by about 5.066 cubic feet.

B3.

The intercept says predicted mean volume is (-36.943) cubic feet when girth is 0 inches; this is usually not practically meaningful for real trees but is the model's mathematical intercept.

B4.

- At $\text{Girth} = 10$: $\widehat{Volume} = -36.943 + 5.066(10) = 13.717$
 - At $\text{Girth} = 16$: $\widehat{Volume} = -36.943 + 5.066(16) = 44.113$
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Part C

- C1: **b**
 - C2: **a**
 - C3: **b**
 - C4: **b**
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Part D

D1.

$$\widehat{Temp} = 96.872 - 1.874(\text{Wind})$$

D2.

- At Wind = 8: $\hat{T}_{emp} = 96.872 - 1.874(8) = 81.880$
 - At Wind = 15: $\hat{T}_{emp} = 96.872 - 1.874(15) = 68.762$
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Part E

Template language to model in each answer:

- **Mean-response confidence interval:** “I am 95% confident that the **mean** value of [response variable + units] is between (L) and (U) when [explanatory variable] = [value + units].”
- **Individual prediction interval:** “I predict with 95% confidence that the [response variable + units] for **one individual** is between (L) and (U) when [explanatory variable] = [value + units].”

E1. Example 1 (trees), Girth = 10

- Confidence interval (11.2, 16.3): We are 95% confident the **mean** tree volume for all trees with girth 10 inches is between 11.2 and 16.3 cubic feet.
- Prediction interval (3.5, 23.9): We predict one **individual** tree with girth 10 inches will have volume between 3.5 and 23.9 cubic feet with 95% prediction confidence.

E2. Example 1 (trees), Girth = 16

- Confidence interval (41.0, 47.2): We are 95% confident the **mean** volume for all trees with girth 16 inches is between 41.0 and 47.2 cubic feet.
- Prediction interval (31.5, 56.6): We predict one **individual** tree with girth 16 inches will have volume between 31.5 and 56.6 cubic feet with 95% prediction confidence.

E3. Example 2 (wind vs temperature), Wind = 8

- Confidence interval (79.8, 84.0): We are 95% confident the **mean** temperature on days with wind speed 8 mph is between 79.8 and 84.0 °F.
- Prediction interval (69.2, 94.5): We predict one **individual** day with wind speed 8 mph will have temperature between 69.2 and 94.5 °F with 95% prediction confidence.

E4. Example 2 (wind vs temperature), Wind = 15

- Confidence interval (66.4, 71.1): We are 95% confident the **mean** temperature on days with wind speed 15 mph is between 66.4 and 71.1 °F.
- Prediction interval (55.7, 82.0): We predict one **individual** day with wind speed 15 mph will have temperature between 55.7 and 82.0 °F with 95% prediction confidence.